LIBBY — PRECLINICAL HORIZON

hgsoc-tuboovarian-stage3c-r0-hrd-pending-idyq

Preclinical recommendations — forward-looking horizon scan

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Preclinical recommendations

A forward-looking horizon scan of candidate drugs, compounds, and treatment strategies that are earlier than clinical development, or not yet developed, but plausibly target this patient's stated features. These are research directions, not enrollable options or treatment recommendations.

1. PARG inhibitor (PDD00017273-class) (small molecule)

Target: genomic instability / HRD (BRCA-proficient) **Stage:** In vivo (animal) **Evidence strength:** Emerging

Strongest preclinical case in the set: a replication-stress synthetic lethality that fits a BRCA-proficient, genomically unstable tumor, gated on a response biomarker that is not yet clinical.

This is the genomically unstable, BRCA-proficient HGSOC background where PARG inhibition has been most active, and it works through a different lesion than PARP trapping, so a genomic-instability rationale carries it even without a BRCA gate.

Mechanism. Blocking PARG traps poly(ADP-ribose) on chromatin during replication, collapsing already-stressed forks and leaving single-stranded DNA gaps that kill the cell.

Translatability. Moderate: selective killing in ovarian lines and HGSOC xenografts with a candidate ssDNA-gap biomarker, though in vivo depth is still limited.

Developability. Tool probe only. A clinical-grade PARG inhibitor and a validated response assay would both be needed first.

Counter-productive MoA. Low (Acts on a distinct lesion from PARP trapping, so little mechanism-level conflict with the goal)

Key risks.

- No validated patient-selection assay for PARG dependence exists yet
- Replication-stress agents can carry marrow toxicity, which runs against this patient's myelosuppression veto
- Most data are cell-line based, so response durability in vivo is uncertain

Open questions.

- Does this tumor carry the ssDNA-gap / low fork-protection signature that predicts PARG sensitivity?
- Would a clinical-grade PARG inhibitor reproduce the tool-compound effect?

Key references. pmid:30889383; pmid:39546575

2. cGAS-STING agonism exploiting chromosomal instability, paired with the IL-12 axis (conceptual strategy)

Target: genomic instability / micronuclei + IL-12 microenvironment **Stage:** Conceptual **Evidence strength:** Speculative

A complementary immune lever for an MSS, low-TMB tumor already on IL-12 therapy, but the specific combination is untested and carries real inflammatory-toxicity and immunosuppression risk.

The tumor's genomic instability is the substrate cGAS-STING reads, and the patient is already on IL-12 immunogene therapy, so an agonist could be complementary rather than redundant in an MSS, low-TMB tumor that single-agent checkpoint blockade is unlikely to help.

Mechanism. Chromosomally unstable tumors spill DNA into the cytosol through micronuclei; a STING agonist pushes that cGAS-STING danger signal deliberately, and layered onto the patient's IL-12 plasmid the two feed the innate-to-adaptive handoff from different angles.

Translatability. Low: no data for this specific STING-plus-IL-12 pairing; supported only by separate CIN-immunity and ovarian manganese-plus-IL-12 work.

Developability. STING agonists remain hard to deliver and dose safely. Sequencing one against active IL-12 therapy would need model work before any patient test.

Counter-productive MoA. Moderate (Chronic STING signaling can flip to immunosuppressive or pro-metastatic, working against the immune goal)

Key risks.

- Systemic STING activation can drive cytokine toxicity that compounds an already active IL-12 therapy
- Chronic STING signaling can turn immunosuppressive or pro-metastatic in some settings
- No data exist for this exact combination

Open questions.

- Does this tumor show micronuclei / cytosolic-DNA signaling that an agonist could amplify?
- How would a STING agonist be sequenced safely against the on-study IL-12 plasmid?

Key references. pmid:29342134; pmid:32820094; pmid:38175694

3. HER2-TKI priming to raise ADC delivery in HER2-low tumors (TKI + HER2 ADC) (combination)

Target: HER2-low expression (IHC 1+) **Stage:** In vivo (animal) **Evidence strength:** Emerging

The best-evidenced and most buildable of the HER2-low ideas, but every supporting study used HER2-amplified disease, so whether priming lifts an IHC 1+ ovarian tumor into a responsive range is the whole question.

The hard part of treating a HER2-low IHC 1+ tumor with a HER2 ADC is that there is little antigen to bind and internalize. Priming with a TKI to nudge surface HER2 up, or to speed payload delivery, is one way to widen that narrow window. Of the three HER2-low ideas here it has the most in-vivo support and uses drugs that already exist.

Mechanism. HER2 tyrosine-kinase inhibitors change how much receptor sits on the cell surface and how fast it cycles. Reversible inhibitors like lapatinib let HER2 pile up at the membrane; irreversible ones like pyrotinib speed its internalization. Either route has raised ADC uptake in models, so a TKI could briefly make a low-HER2 tumor more visible or more permeable to a HER2 ADC.

Translatability. Low / cross-tumor only: in-vivo proof that pyrotinib raises T-DXd uptake comes from gastric and breast HER2-amplified models, never HER2-low or ovarian.

Developability. Closest of the HER2 set: both drug classes are approved, so a priming-then-ADC schedule could be tested without new chemistry. The sequence and timing are unworked.

Counter-productive MoA. Moderate (The wrong TKI or schedule lowers surface HER2 and cuts ADC delivery instead of raising it)

Key risks.

- Reversible and irreversible TKIs move HER2 in opposite directions, so the wrong agent or schedule could cut rather than raise ADC delivery
- All data are in HER2-amplified models, so the absolute lift in a 1+ tumor may be too small to matter
- A TKI adds its own diarrhea and rash on top of the ADC, which sits against this patient's low-burden preference

Open questions.

- Does TKI priming raise surface HER2 enough in a 1+ tumor to change ADC response?
- Which TKI and what schedule, given reversible and irreversible agents push HER2 in opposite directions?

Key references. pmid:19060928; pmid:38386239

4. Oral ER degrader (vepdegestrant / ARV-471) repurposed for ER-positive ovarian cancer (repurposed drug)

Target: ER-positive (PR-negative) expression **Stage:** Conceptual **Evidence strength:** Speculative

The most readily sourced idea, an oral ER degrader matching the patient's preference, but with no ovarian evidence and a PR-negative discount on expected benefit.

This HGSOC is ER-positive and the patient prefers low-burden oral therapy, which fits an oral ER degrader. It is a different mechanism from the aromatase inhibition already on the case, so it is a distinct lever on the same feature rather than a repeat, though PR-negativity tempers the expected magnitude.

Mechanism. An oral ER PROTAC binds the estrogen receptor and an E3 ligase at once, tagging ER for degradation rather than only blocking ligand binding, which removes receptor protein more completely than an aromatase inhibitor that only lowers estrogen supply.

Translatability. Low: efficacy data are entirely in ER-positive breast models, there is no ovarian package, and PR-negativity discounts the expected benefit.

Developability. Closest to reach: the molecule is clinically mature in breast cancer, but no ovarian preclinical package supports a repurposing trial yet.

Counter-productive MoA. Low (Deeper ER suppression is on-goal in ER-positive disease; main risk is simply limited efficacy)

Key risks.

- No ovarian-specific efficacy data
- ER positivity predicts response inconsistently in HGSOC, and PR-negativity is a further discount
- Benefit may be modest next to the endocrine options already in play

Open questions.

- Does ER degradation outperform aromatase inhibition in ER-positive, PR-negative HGSOC?
- Is the PR-negative subset likely to derive any meaningful endocrine benefit?

Key references. pmid:38819400

5. RAD51 inhibitor (chemical probes B02 / B02-isomer) (small molecule)

Target: genomic instability / HRD (BRCA-proficient) **Stage:** In vitro **Evidence strength:** Speculative

A non-genetic route to PARP sensitization in BRCA-wild-type disease, held back by weak chemical probes and the absence of any clinical-grade inhibitor.

A genomically unstable but BRCA-proficient tumor still has functional RAD51, so chemically dialing down HR is one route to mimic the deficiency PARP inhibitors exploit. The draw is a non-genetic path to PARP sensitization in BRCA-wild-type disease.

Mechanism. RAD51 drives the strand-invasion step of homologous recombination; B02-class probes block RAD51 filament formation to suppress HR and force an HR-deficient state pharmacologically, to be paired with a PARP-trapping or DNA-damaging agent.

Translatability. Low: cross-tumor cell-line data only, current probes are weak, and no drug-like RAD51 inhibitor exists.

Developability. Bench reagents with no clinical-grade successor. A genuine drug-like RAD51 inhibitor would have to be built first.

Counter-productive MoA. Low (On-mechanism HR suppression supports the goal; main hazard is normal-tissue genotoxicity, an AE rather than a goal conflict)

Key risks.

- Systemic HR suppression could be genotoxic to normal tissue
- Probe potency and selectivity are weak
- The CYT-0851 story shows how a RAD51-inhibitor phenotype can come from an off-target mechanism

Open questions.

- Can a selective, drug-like RAD51 inhibitor be developed at all?
- Would pharmacologic HR suppression sensitize this tumor without unacceptable normal-tissue toxicity?

Key references. pmid:21428443; pmid:34208492

6. PI3K-alpha targeted degradation (PROTAC) concept (PROTAC / degrader)

Target: somatic PIK3CA amplification **Stage:** In vivo (animal) **Evidence strength:** Speculative

A modality-distinct answer to PIK3CA amplification rather than mutation, but the amplification-to-degrader rationale is reasoned, not yet shown.

This tumor carries a focal PIK3CA amplification, not a hotspot mutation, whereas the catalytic inhibitors already on the case were built for mutant or pathway-active disease. A degrader is a mechanistically distinct option whose protein-lowering action maps more naturally onto an amplification lesion.

Mechanism. A PROTAC removes the PI3K-alpha protein rather than only blocking its kinase pocket, lowering both catalytic and scaffolding output, which could in principle counter the gene-dosage effect of an amplified, overexpressed PIK3CA that catalytic inhibitors leave untouched.

Translatability. Low: PI3K degraders are active in animal models, but none was tested in the amplification setting this rec rests on.

Developability. Early discovery compounds only. A clinical-grade degrader and amplification-specific validation are both missing.

Counter-productive MoA. Low (Lowering PI3K-alpha protein is on-goal for an amplification; metabolic toxicity is an AE, not a goal conflict)

Key risks.

- The amplification-to-degrader rationale is unproven; most data concern mutant PI3K-alpha
- On-target hyperglycemia tracks the whole PI3K-alpha axis
- No clinical-grade molecule exists

Open questions.

- Does protein degradation outperform catalytic inhibition specifically in PIK3CA-amplified models?
- Can a PI3K-alpha degrader spare normal metabolic tissue enough to be tolerable?

Key references. pmid:40518729; doi:10.1158/2159-8290.CD-21-1411

7. HER2 PROTAC degrader (tucatinib-based tool compound CH7C4) (PROTAC / degrader)

Target: HER2-low expression (IHC 1+) **Stage:** In vivo (animal) **Evidence strength:** Speculative

Mechanistically this fits HER2-amplified disease far better than the IHC 1+ tumor in front of us, so it reads as a HER2 direction to track rather than a worked-out rationale for HER2-low.

A degrader is the one HER2-directed modality not already on the case as an ADC, so it is worth logging. The honest problem is that the proof sits in HER2-amplified breast cells where HER2 is the driver; in a HER2-low tumor the receptor is unlikely to be what the cell depends on, so removing it may do little.

Mechanism. A PROTAC links a HER2 binder, here built on the kinase inhibitor tucatinib, to an E3-ligase recruiter and tags the receptor for proteasomal destruction rather than only blocking its kinase pocket. Pulling HER2 out cuts both its signaling and its scaffolding. The reported compound degrades HER2 with a DC50 of 69 nM and near-complete degradation.

Translatability. Low / cross-tumor only: all degradation and tumor-growth data come from BT-474 HER2-amplified breast cells and xenografts, not HER2-low or ovarian disease.

Developability. Far from a patient: a laboratory compound with no HER2 degrader yet in any oncology trial. It would need a drug-like molecule plus evidence that a HER2-low tumor depends on HER2 at all.

Counter-productive MoA. Low (No active goal conflict; the real hazard is futility if the tumor does not depend on HER2)

Key risks.

- HER2-low tumors are generally not HER2-driven, so degrading the receptor may have little antitumor effect
- All efficacy data are in HER2-amplified breast models
- No clinical-grade HER2 degrader exists

Open questions.

- Does a HER2-low tumor depend on HER2 enough for degradation to matter?
- Could a drug-like HER2 degrader be built from this tool compound?

Key references. pmid:36208507

8. MAP2K7 (MKK7) paralog-dependency synthetic-lethal concept (conceptual strategy)

Target: somatic MAP2K4 deletion **Stage:** Conceptual **Evidence strength:** Speculative

The only named angle on MAP2K4 loss, but a pure hypothesis several validation steps from any tool compound or patient.

This tumor carries a focal somatic MAP2K4 deletion and paralog dependency is a recurring theme in functional-genomics screens, so MKK7 is a logical place to look for a matched vulnerability where no targeted therapy for MAP2K4 loss exists today.

Mechanism. MAP2K4 and MAP2K7 are paralogous kinases that both feed JNK stress signaling, so a MAP2K4-deleted tumor may lean harder on MAP2K7, the setup behind many paralog synthetic-lethal pairs, making MKK7 a candidate vulnerability.

Translatability. Low: a paralog-dependency hypothesis with no experimental support in MAP2K4-deleted ovarian cancer and no selective tool compound.

Developability. Furthest from a patient: no MKK7-selective inhibitor of clinical quality, and the synthetic-lethal link itself would need demonstrating in isogenic models first.

Counter-productive MoA. Moderate (JNK can be tumor-suppressive, so blocking MKK7 could promote rather than restrain growth)

Key risks.

- The paralog dependency is assumed, not shown, for MAP2K4 loss
- JNK signaling is context-dependent and can be tumor-suppressive, so MKK7 inhibition could backfire
- No tool compound with adequate selectivity exists

Open questions.

- Is MAP2K4 loss actually synthetic-lethal with MAP2K7 in isogenic ovarian models?
- Could a selective MKK7 inhibitor be made without hitting tumor-suppressive JNK functions?

9. High-avidity HER2-low-directed T-cell engager / CAR concept (conceptual strategy)

Target: HER2-low expression (IHC 1+) **Stage:** Conceptual **Evidence strength:** Speculative

Named mainly to document why cell therapy against a HER2-low tumor is hard: the safety problem would have to be solved before a molecule could even be made.

Cell therapy is the one HER2 modality with no representation on this case, so it is worth naming. But the field is moving the other way: low-affinity HER2 CARs are being built specifically to ignore HER2-low cells, because that level shows up on normal heart and gut. A receptor sensitive enough to hit this 1+ tumor would, by the same logic, risk those tissues.

Mechanism. A T-cell-redirecting therapy, a CAR or a bispecific engager, points cytotoxic T cells at surface HER2 without a chemotherapy payload. To act on an IHC 1+ tumor it would have to fire at very sparse antigen, which means a high-avidity receptor rather than the affinity-tuned designs now in favor.

Translatability. Low / none: published cell-therapy work is engineered to spare HER2-low, used as a normal-tissue surrogate, so the construct this would need does not exist.

Developability. Furthest from a patient of the HER2 set: no construct is designed to engage HER2-low, and building one runs straight into the on-target off-tumor toxicity the affinity-tuning work was created to avoid.

Counter-productive MoA. High (A receptor sensitive enough to hit an IHC 1+ tumor would also attack HER2-low heart and gut)

Key risks.

- On-target off-tumor toxicity against HER2-low normal epithelium in heart and GI is the central barrier
- The field is deliberately engineering away from HER2-low, so the high-sensitivity construct this needs does not exist
- No ovarian or HER2-low efficacy data of any kind

Open questions.

- Could a construct tell an IHC 1+ tumor apart from HER2-low normal heart and gut at all?

- Is there any therapeutic window for HER2-low T-cell redirection?

Key references. pmid:38325903